

// PHP 8.2 · LARAVEL 11 · MYSQL · DOCKER · PSR-12

Project Manager

// enterprise admin dashboard

A Laravel 11 admin panel with multi-role access control, product approval workflows, rich-text CRUD, full-text search, and a clean Repository + Service architecture — demonstrating enterprise PHP patterns from day one.

3

USER ROLES

SOLID

PRINCIPLES

PSR-12

CODE STYLE

4

DOCKER SERVICES

// 01 - TECHNOLOGY

Technology Stack

LANGUAGE

PHP 8.2

strict_types, named args,
match expressions

FRAMEWORK

Laravel 11

Streamlined
bootstrap/app.php, no
Kernel.php

DATABASE

MySQL 8

Eloquent ORM, FULLTEXT
index on products

AUTH

Laravel Auth

Password login, bcrypt hashed,
rate-limited

FRONTEND

Blade + Bootstrap 5

Quill rich-text editor for
product descriptions

INFRASTRUCTURE

Docker Compose

Nginx · PHP-FPM · MySQL ·
MailHog in containers

CODE QUALITY

EMAIL

PSR-4 + PSR-12

Laravel Pint linter (make lint)

Laravel Mail

Product status emails, caught by MailHog locally

// 02 - ARCHITECTURE

Clean 4-Layer Architecture

HTTP Request

Browser → Nginx → public/index.php



Controller

HTTP adapter only — validates input, returns response. Zero query logic.



Service Layer

Business rules live here — ownership checks, approval transitions, role guards.



Repository

All database queries. Implements ProductRepositoryInterface contract.



Eloquent Model → MySQL

Product, User — parameterised queries, soft deletes, eager loading



architecture.php

```
// Request hits ProductController@store (POST /products)
```

```
ProductRequest    → authorize() checks role, prepareForValidation() strips XSS
```

```
ProductController → calls service, returns redirect. No query logic here.
```

```
ProductService    → stamps created_by, sets status='pending', checks canModify()
```

```
ProductRepository    → Product::create($data) – the only place queries are written
MySQL                → row inserted, id returned

// Email notification (admin alerted to new pending product)
ProductService → Mail::to(admin)->send(new ProductSubmittedMail($product))
```

// 03 – FEATURES

What Each Role Can Do

Admin — Full Control

- ✓ Approve or reject any pending product
- ✓ Edit or soft-delete any product regardless of creator
- ✓ Create and manage user accounts
- ✓ Assign roles (admin / product_manager / customer)
- ✓ View all orders across all customers
- ✓ Update order status (triggers customer email)
- ✓ Email sent to manager on approval/rejection

Product Manager

- ✓ Create products with Quill rich-text description
- ✓ Set price, category, stock quantity
- ✓ Products enter pending state automatically
- ✓ Edit and soft-delete own products only
- ✓ Full-text search across title + description
- ✓ Pagination on product list (12 per page)
- ✓ Receive email notification on approve/reject

Security Built-In

- ✓ XSS sanitisation on all rich-text input (strip_tags allow-list)
- ✓ CSRF token on every state-changing form

Developer Experience

- ✓ `bash deploy.sh` — full setup in one command
- ✓ Makefile: make lint, make test, make fresh, make shell

✓ Rate-limited login — 5 attempts then
lockout

✓ Role checked at route level AND in service
layer

✓ Ownership check — manager can't edit
others' products

✓ Soft deletes — products recoverable after
deletion

✓ MailHog UI at localhost:8025 — see all
emails instantly

✓ Laravel Pint enforces PSR-12 style
automatically

✓ PHPUnit test suite wired up (make test)

// 04 — KEY WORKFLOWS

How Core Processes Work

✓ Product Approval Flow

- 1 Manager creates product**
Status automatically set to pending in ProductService
- 2 Admin receives email**
ProductSubmittedMail dispatched to all admins
- 3 Admin reviews & decides**
Clicks Approve or Reject from the dashboard
- 4 Manager notified**
ProductStatusMail sent to the product's creator
- 5 Approved = visible in shop**
ShopController query uses scopeApproved() automatically

🔑 Password Authentication Flow

- 1 User submits credentials**
LoginRequest validates format
- 2 Rate limiter checks attempts**
5 failures per email+IP = logout + Lockout event
- 3 Auth::attempt() runs**
Bcrypt comparison, 12 rounds
- 4 Session regenerated**
Prevents session fixation attack
- 5 Role-appropriate redirect**
Admin → /dashboard, else → /shop

// 05 - SECURITY

Security Measures

THREAT	HOW IT IS STOPPED
SQL Injection	Eloquent parameterised query builder throughout. No raw SQL built from user input.
Stored XSS	ProductRequest::prepareForValidation() calls strip_tags() with a safe-tag allow-list before any data is validated or saved.
CSRF	@csrf on every POST/PATCH/DELETE form. VerifyCsrfToken middleware rejects missing tokens.

THREAT	HOW IT IS STOPPED
Brute Force Login	RateLimiter — 5 attempts per email+IP, then throttle. Lockout event fired.
Privilege Escalation	CheckRole middleware blocks at route level. ProductService::canModify() enforces ownership at object level. abort(403) on failure.
Mass Assignment	\$fillable whitelist on every model. Attackers cannot inject role=admin via a crafted form submission.
Session Fixation	\$request->session()->regenerate() on login. invalidate() + regenerateToken() on logout.
Sensitive Files	Nginx denies /dotfile requests. .env is gitignored. Docker image never exposes .env.

// 06 – GETTING STARTED

Run Locally in 3 Commands

BASH – Docker setup

Clone and enter

```
git clone https://github.com/vasant24konar/Project-Manager.git && cd Project-Manager
```

Copy environment file (no edits needed for local Docker)

```
cp .env.example .env
```

Bootstrap everything (builds images, migrates DB, seeds data)

```
bash deploy.sh
```

App running at

```
http://localhost:8080      # Admin dashboard
```

```
http://localhost:8025     # MailHog – approval notification emails
```

ROLE	EMAIL	PASSWORD	ACCESS
Admin	admin@example.com	Admin@1234	Full dashboard — approve/reject/manage users
Product Manager	manager@example.com	Manager@1234	Create products, see own pending status

// 07 - REPOSITORY

Source Code



Project Manager

Laravel 11 admin dashboard — RBAC, approval workflow, rich-text CRUD, full-text search, Docker.

github.com/vasant24konar/Project-Manager

Project Manager — Built with Laravel 11 • PHP 8.2 • MySQL 8 • Docker

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